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An Efficient and Transparent Financial Transaction System using Decentralized Finance (DeFi) based on Blockchain Technology

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Abstract— ***A cutting-edge financial innovation that poses a challenge to the existing centralised banking system is decentralised finance, or DeFi. DeFi enables the trend of peer-to-peer transactions which eliminates the charges that banks and other financial service providers levy on their services. Anyone with internet access can employ software that tracks and verifies financial transactions in distributed databases of finance to, trade, lend and borrow money. That is why the implementation of smart contracts allows DeFi to create more transparent, open, and harmonically integrated protocols for modern financial services.. Decentralisation carries the risk of undermining existing accountability frameworks as well as the effectiveness of current financial regulatory and enforcement mechanisms. Simultaneously, in decentralised financial services it can be seen that value chains, reconcentration occurs in a different (perhaps less regulated, less apparent, and less transparent) value chain component. The primary objective of DeFi, decentralisation, cannot be achieved without regulation; in fact, regulation is necessary for DeFi to function. The proposed paper presents an innovative framework for decentralised finance that successfully addresses significant issues in the current financial ecosystem such as transaction process inefficiencies, vulnerabilities in security, and prohibited accessibility. The proposed approach not only increases user confidence and privacy, but also guarantees that financial services are more accessible to marginalised populations by integrating cutting-edge cryptography techniques with creative governance strategies. The experimental investigations' results indicate that the suggested framework can be effectively established in banks to gain greater accessibility, better transparency, and more flexibility for the users.***

Keywords— Block Chain, DeFi, Cryptocurrency, Smart Contract

I. Introduction

Staking, or Decentralized Finance (DeFi) is the practice of a user locking or holding their money in a cryptocurrency wallet in order to take part in the upkeep of a blockchain system that is based on proof-of-stake (PoS)[1]. A type of blockchain consensus algorithms known as PoS protocols chooses validators based on how much of the corresponding cryptocurrency they own. Stated differently, it is the process of staking cryptocurrency to be used as collateral by PoS blockchains in order to accomplish certain goals, such as verifying transactions and collecting interest or incentives in the form of freshly created tokens[2]. It works similarly to blockchain mining in that it rewards users for participating and helps the network reach consensus. The number of coins "locked" in a wallet determines who has the authority to validate transactions[3]. Stakers are motivated to decide which block to add to a blockchain or to be the next miner on a proof-of-work (PoW) network. Thus, the blockchain consensus process that underpins mining and staking is what really distinguishes the two practices[4]. PoW is implemented by mining, most famously in bitcoin (BTC). While the majority of cryptocurrencies employ staking, PoS blockchain platforms, like Ethereum 2, mark an important turning point in the development of blockchain technology. They have quick transaction times and are scalable. Ethereum has switched from a PoW to a PoS system[5]. The consensus protocol of choice is staking since PoW mining is an unsustainable strategy. Primarily, PoW necessitates specialized technology (such as a GPU) and is energy-intensive. With a 99% energy usage reduction, staking is generally seen as being more ecologically sustainable. In its most limited sense, DeFi staking is the process of enclosing cryptocurrency assets in a smart contract in return for the opportunity to serve as a validator on a Layer 1 blockchain or a DeFi protocol, thereafter the stakeholder receives incentives for carrying out the requisite tasks[6]. More widely, the phrase "DeFi staking" is frequently used to refer to any DeFi activity that calls for the temporary commitment of cryptocurrency assets[7].

DeFi has many important benefits in comparison to centralized finance systems since there are no third parties such as banks and brokers. For example, due to the implementation of smart contracts and adoption of Blockchain technology , DeFi aims at allowing users to make transactions directly without having to involve third parties which save time as well as cost in the process. It has also been stated that due to the absence of intermediaries, not only are the transaction costs lowered but there are also fewer delays and risk of human error. Furthermore, DeFi is more robust because its platforms are decentralized so there is no single entity controlling them and therefore there is minimal interference by the authorities or any institutions. This means that the users do not lose complete control of their funds and financial activities.Inclusiveness and acceptability are also other areas which are well catered for when it comes to DeFi rebuilding. A person from any part of the world with Internet connectivity can access DeFi platforms without considering his/her geographical area, credit score or wealth status etc. This expands financial services like lending, borrowing or investing to populations that are frequently left out as a result of the traditionally centralized financial systems. Aside from this, more

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transparency can be achieved since every transaction by the DeFi is done on the blockchain and therefore is considered an open ledger. Lastly, the remoteness of DeFi due to the use of smart contracts enhances speed of development leading to new financial products and services which do not require any type of supervision.

The proposed DeFi architecture incorporates a number of interrelated principles that serve to both secure and efficiently execute economic transactions primarily based on blockchain and smart contracts. In the context of a DeFi application, smart contracts can be described as computer programs that enable self-executing contracts and which have limited the terms of the agreement to code. These contracts only implement the agreed terms after the terms laid in the contracts have been fulfilled thus nixing the need for middlemen. This reduces likely hood of manipulation, fraudulent practices and or interference by a third party since contracts are executed within a blockchain. DeFi is based on a decentralized consensus on the blockchain which distributes control over many nodes. Therefore, the desired autonomy of institutions shall be achieved without the requirement to reference a single source, which would created a single point of failure and make the system more vulnerable to attacks or censorship. These proposed DeFi platforms utilize the advanced features of Homomorphic Encryption of the credential and assets of its users. This allowed for computation on data without compromising on privacy and confidentiality can be used side by side with public private key encryption with only the funds' control and accessibility granted to the true owner while transactions are checked and the ecosystem maintained through Practical Byzantine Fault Tolerance (PBFT) or Delegated Proof of Stake (DPoS) .

The proposed approach will greatly improve financial inclusion for underprivileged groups by offering access to financial services without the need of traditional banking infrastructure. Individuals in underdeveloped or rural locations can use DeFi systems with just an internet connection and a digital wallet, eliminating traditional banking hurdles like minimum balances and required verification. This access allows people to engage in a variety of financial activities, including saving, borrowing, and trading, without the constraints imposed by traditional financial institutions. Furthermore, DeFi frequently can have cheaper fees than traditional banking, making it more accessible to consumers who may struggle with high transaction expenses.It, therefore, caters for everyone including those who would have found it quite challenging owing to high transaction charges. Furthermore, DeFi frequently can have cheaper fees than traditional banking, making it more accessible to consumers who may struggle with high transaction expenses. Furthermore, DeFi encourages financial autonomy and empowerment by allowing individuals to keep ownership of their assets without depending on middlemen. This is especially important for impoverished people who may reject centralised financial institutions as a result of exclusion or mismanagement.

Many people and organizations are unable to utilize traditional financial systems due to their inefficiencies, lack of transparency, and entrance obstacles[8]. DeFi leverages the inherent properties of blockchain technology to create a decentralised financial services ecosystem accessible to all users having an internet connection, irrespective of location or socioeconomic status. Fundamentally, DeFi makes use of blockchain networks to enable peer-to-peer transactions, doing away with the need for middlemen like banks or other financial organizations[9]. DeFi apps automate a range of financial transactions with a high level of security and transparency, including borrowing, lending, trading, and asset management. They do this by utilizing smart contracts, which are self-executing code that is placed on the blockchain.

Traditional DeFi has a number of severe issues and problems, including security flaws, user mistakes, and regulatory uncertainties. Security breaches caused by vulnerabilities in smart contracts or exploits can result in significant financial losses for users, as proven by high-profile attacks in the DeFi area. Furthermore, the intricacy of DeFi protocols frequently leads in user mistakes, such as transmitting payments to the incorrect address or losing access to private keys. Regulatory uncertainty also creates dangers, since governments may apply limits that jeopardise the sustainability of DeFi initiatives or cause abrupt shutdowns. To overcome these issues, the proposed DeFi system incorporated a variety of novel solutions aimed at improving security and usability. This include rigorous smart contract audits, formal verification methods, and user-friendly interfaces that help people navigate the complexity of maintaining their assets and conducting transactions.

The use of smart contracts into the proposed DeFi system greatly adds to the automation of financial operations by allowing self-executing agreements that work under predetermined criteria encoded into code. Smart contracts expedite financing, borrowing, trading, and insurance operations by automatically facilitating, verifying, or enforcing transaction conditions without the need for middlemen.For example, in a loan scenario, a smart contract may immediately transfer cash to a borrower after collateral is received, and it can also secure repayment by deducting payments directly from the borrower's account on the agreed-upon timetable. This automation saves time and effort when conducting financial transactions, decreases human error, and improves overall financial service efficiency. The implications of this automation are significant. For example, it improves accessibility by letting customers to interact with financial services at any time of day or night, without regard for typical banking hours or intermediary availability. By providing people with continuous access to financial resources, this may empower impoverished populations. Second, automation increases transparency and confidence by logging every transaction on an open blockchain, which enables users to independently confirm actions without relying on a central authority. Furthermore, the absence of middlemen might result in cost savings since it lowers transaction fees and operational expenses connected with traditional financial systems.

In this work, the use DeFi, a blockchain-based financial transaction system is analyzed to create a transparent and efficient system. The work is also explored on the essential elements, factors, and best practices associated with constructing such a system, covering everything from user interface design and regulatory compliance to smart contract

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creation and security assessments. Organizations and developers may seize new chances to transform the financial environment and make it more robust, inclusive, and accessible for everyone by utilizing DeFi and blockchain technology.The primary goals of the suggested paper are as follows.

- **Improving Financial Inclusion:** Provide underprivileged groups, such as those residing in areas with inadequate financial infrastructure or those without access to typical banking services, with means of obtaining financial services . .
  - **Lowering Intermediary Cost:** By relying less on intermediaries like banks or other financial organizations, you may cut down on the overhead and transaction fees that come with using traditional financial systems.
- **Increasing Transaction Efficiency:** Improve the speed and efficiency of financial transactions by leveraging blockchain technology and smart contracts to automate processes and eliminate manual intervention.
- **Improve Accountability and Transparency:** Using a public blockchain ledger for real-time audits and verification, proposed system will enhance transparency and accountability in financial transactions.

II. LITERATURE REVIEW

This is a survey of the literature that highlights important studies and academic papers about deploying a transparent and effective financial transaction system with DeFi built on blockchain technology.A thorough description of DeFi is given in the paper[10], which also addresses its problems, future potential, ecosystem components, and historical growth. It covers a range of DeFi applications, including as asset management protocols, stablecoins, lending systems, and decentralized exchanges.Leading blockchain technology startup ConsenSys provides a thorough review of DeFi and how it can upend conventional finance[11]. The paper examines the advantages of DeFi, including its composability, accessibility, and transparency. It addresses how blockchain technology makes DeFi possible and examines how it affects innovation, legislation, and financial inclusion. The study article [12] investigates how DeFi may advance financial inclusion by giving marginalized groups access to financial services. It looks at case studies of DeFi initiatives that tackle certain issues related to financial inclusion and talks about the advantages and disadvantages of adopting DeFi.The potential, dangers, and regulatory ramifications of DeFi are examined in this research. It addresses the difficulties with security, scalability, and regulatory compliance in addition to the possible advantages of DeFi, such as enhanced efficiency, transparency, and accessibility. An overview of DeFi protocols and the supporting technologies—such as Ethereum smart contracts, decentralized exchanges, and lending platforms—is given in the article [13]. It explores the ideas of decentralization, transparency, and trust lessness that guide DeFi and examines how they affect conventional finance. The advent of decentralized finance and its potential to transform financial markets are examined in the research paper [14]. It examines the benefits of DeFi, including lower counterparty risk, more liquidity, and automated execution, and assesses how it may affect established financial institutions and regulatory frameworks.. In order to improve the scalability of DeFi platforms, the study presented in [15] concentrated on Layer 2 scaling solutions for blockchain networks. These solutions, such as Optimistic Rollups and zkRollups, process transactions off-chain and settle them on-chain. By boosting throughput and drastically lowering transaction costs, these technologies can improve DeFi's usability and effectiveness.The use of methods like homomorphic encryption, multi-party computation (MPC), and zero-knowledge proofs (ZKPs) to increase privacy and security in DeFi is covered in [16]. The goal of these privacy-enhancing solutions is to safeguard private user information and transactions without compromising blockchain networks' auditability and openness.There is a growing movement to tokenize physical assets including stocks, commodities, and real estate.Tokenization and investment in the DeFi market were discussed in [17]. By allowing tokenized assets to be exchanged and moved across blockchain networks, formerly illiquid assets can now have partial ownership and liquidity. These advancements broaden the definition of DeFi and create additional investment and asset tokenization options. The goal of the study put forward in[18] was to improve the efficacy, inclusiveness, and openness of DAO governance frameworks. DAOs are starting to take center stage as DeFi platforms and protocols' governance systems. Voting on protocol updates, financing proposals, and other governance issues is made possible by DAOs, which facilitate decentralized decision-making and community governance. The project suggested in [19] Give predictions on the future of blockchain technology and DeFi, taking into account new developments, developing trends, and possible disruptions. Prospects for institutional involvement, mainstream acceptance, and integration with established financial infrastructure were also covered[20].These literary sources provide insightful information on the ideas, uses, difficulties, and prospects around blockchain-based DeFi. These materials can be used by scholars, practitioners, and policymakers to further their knowledge of DeFi and improve decentralized financial systems[21].

III. PROPOSED METHODOLOGY

The design of an effective and transparent financial transaction system that uses blockchain-based DeFi consists of many linked layers that enable smooth operation and user engagement. The User Interface (UI) layer is at the forefront and acts as a user's entry point into the system. Users engage with a variety of services, including asset management, trading, lending, and borrowing, using online interfaces or mobile applications. The Backend Services layer, which houses the system's processing power and business logic, is located behind the UI layer. This layer includes off-chain processing services, smart contracts, and decentralized applications (dApps). Encoded in the blockchain, smart contracts guarantee transparent and trustless processes by automating transaction execution and enforcing predetermined rules. APIs also let the UI and backend communicate with each other. The UI layer is at the

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forefront and acts as a user's entry point into the system. Users engage with a variety of services, including asset management, trading, lending, and borrowing, using online interfaces or mobile applications. The Backend Services layer, which houses the system's processing power and business logic, is located behind the UI layer. This layer includes off-chain processing services, smart contracts, and decentralized applications . Encoded in the blockchain, smart contracts guarantee transparent and trustless processes by automating transaction execution and enforcing predetermined rules. APIs also let the UI and backend communicate with each other.

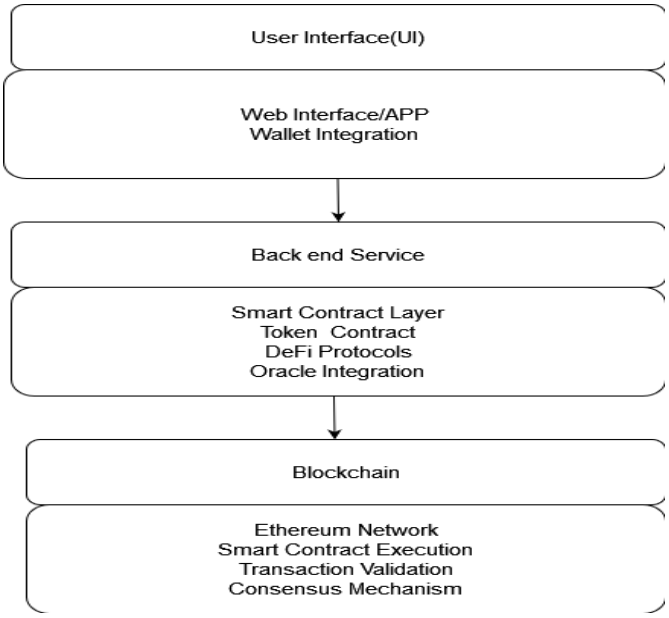

Figure. 1.Proposed Architecture Diagram

The system's fundamental building block, the Blockchain Network, offers a decentralized, unchangeable ledger for keeping track of transactions. The blockchain network, which is made up of a network of nodes, verifies and logs transactions to provide security and transparency. Smart contracts execute transactions in a tamper-proof way by interacting directly with the blockchain. Overall, the architectural diagram shows how to combine oracles, blockchain technology, backend services, and user interfaces to build a decentralized, transparent, and effective financial transaction system that is driven by DeFi. The architecture and flow diagram of the proposed work is shown in Figure 1 and Figure 2.An explanation of each element in the architectural diagram is provided below:

- **User Interface (UI):**

The user interface for the financial transaction system is represented by this layer.Web interfaces, smartphone apps, and other user-facing programs are examples of the UI.Through the user interface (UI), users may carry out a number of tasks, including engaging with decentralized apps (dApps) inside the DeFi ecosystem and accessing their accounts and completing transactions.

- **Backend Services:**

The functionality and business logic of the financial transaction system are included in backend services. This layer consists of things like off-chain processing services, APIs, and smart contracts. Self-executing contracts with pre-established rules inscribed on the blockchain are known as smart contracts. Processes like asset transfers, loan issuing, and decentralized exchange activities are automated by them. Users may engage with the system and obtain data thanks to APIs, which provide communication between the frontend user interface and backend services. Off-chain processing services take care of complicated calculations, data gathering, and system integration—tasks that cannot be carried out on the blockchain.

- **Blockchain Network:**

The financial transaction system's fundamental infrastructure is provided by the blockchain network. It is made up of a decentralized network of nodes that keep a common transaction ledger up to date. Transparency and immutability are ensured by the blockchain network's secure, tamper-proof validation and recording of transactions.

- **Smart Contracts:**

The smart contracts may best be described as automated contracts where contractual provisions are written into the block chain and pre-specified. Smart contracts are program codes running autonomously on the blockchain to enforce specific state and code-based terms of agreements. They oversee the payment process and issue self-execution legal tenders or otherwise known as smart contracts. DeFi employs smart contracts in the running of decentralized exchange, lending and borrowing, token transfers, and liquidity.

- **Oracles**:

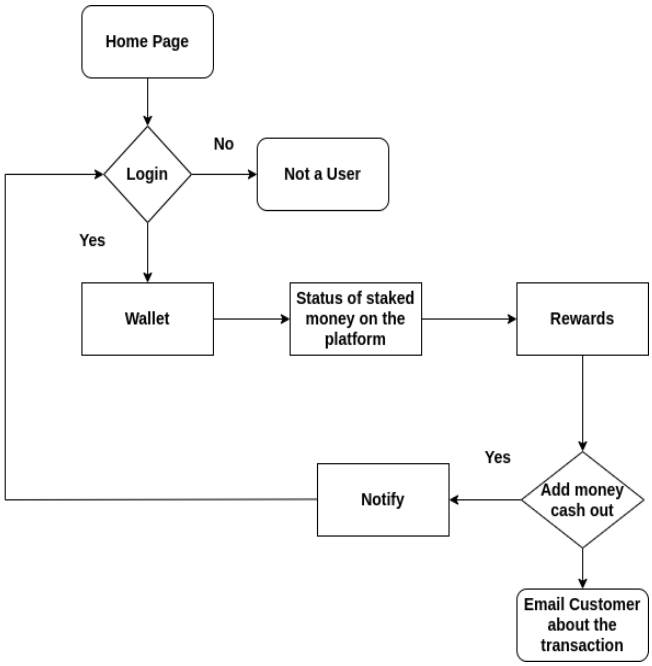

Figure. 2.Flow Diagram of the Proposed Work

Oracles are third-party services or protocols that provide blockchain-based apps access to outside events and real-world data. Oracles are essential to DeFi because they retrieve price feeds, market data, and other information needed to run smart contracts and conduct financial transactions. Oracles support the effective and dependable operation of the financial transaction system by assisting in ensuring the correctness and dependability of data utilized

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within DeFi protocols. Overall, utilizing DeFi based on blockchain technology, the architectural diagram shows how many levels and components interact to create an effective and transparent financial transaction system. This design offers customers a smooth and user-friendly interface for engaging with the system, while also enabling the safe, trustless, and transparent execution of financial transactions.

IV. RESULT AND DISCUSSION

The smart contract in the proposed work is developed using the Ethereum network. Ethereum has a programming language called Solidity that has syntax similar to JavaScript. For creating and testing solidity smart contracts, a variety of development environments are accessible, such as the Remix Truffle framework. The Web3 framework, which enables JavaScript files to interface with the smart contracts, is used to create an intuitive front-end application for the smart contracts. At the heart of the functionality on the front end is React JS. The user's account number and the name of the bank are shown on the Main section's navigation bar, from where they may conduct any cryptocurrency transactions.

The novel smart contract proposed in the proposed paper applies unique "Dynamic Yield Adjustment" technique in the DeFi system, which modifies the yield rates in real-time according to user behavior and market conditions. The proposed solution uses an artificial intelligence (AI) driven algorithm to analyze on-chain data and dynamically change yield rates, in contrast to standard DeFi protocols with fixed interest rates. This method offers customized yield strategies depending on asset volatility and customer risk profiles, which minimizes temporary loss while simultaneously incentivizing liquidity providers to meet market demand. The goal of the proposed paper is to increase user involvement and create a lively community by incorporating gamified aspects like yield farming competitions and awards for active participation. The user interface of the proposed work is shown in Figure 3 below. The experimental configuration that consists of the following elements in order to test and evaluate a novel DeFi application utilizing smart contracts:

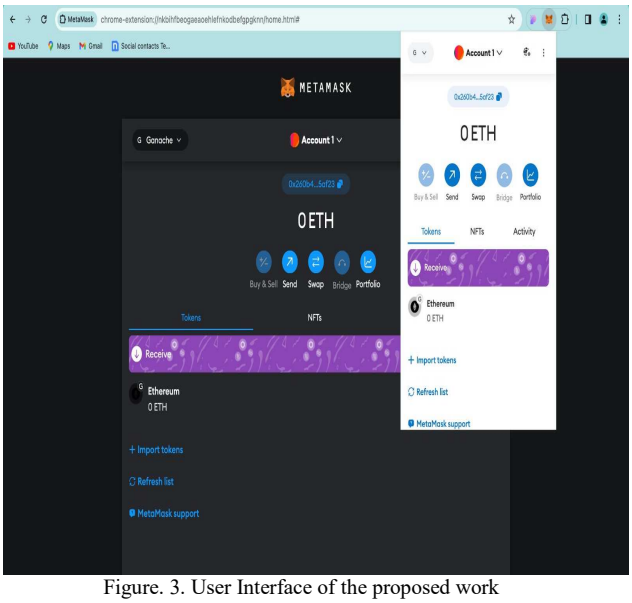

Figure. 3. User Interface of the proposed work

Blockchain Environment: To guarantee real-world circumstances without financial risk, the smart contract is deployed on a testnet using Binance Smart Chain's Testnet **Development Framework:** For the creation, testing, and implementation of smart contracts, the frameworks is used like as Hardhat or Truffle. The testing libraries are used for unit and integration tests, such as Mocha and Chai. **Frontend Interface**: To communicate with the smart contract, intuitive web interface is created with React.js or Vue.js. Ethers.js is used to interface with blockchains.

Data Analysis Tools: To get insights into user behaviour and transaction trends, on-chain data is analyzed using Dune Analytics tool .

TABLE 1.Comparison of Traditional DeFi and Proposed DeFi

Parameter	Traditional DeFi	Proposed Defi
Consensus Mechanism	Predominantly Proof of Work (PoW) or Proof of Stake (PoS)	Delegated Proof of Stake (DPoS) or Practical Byzantine Fault Tolerance (PBFT) for enhanced efficiency and security
Data Encryption	Public-key cryptography	zero-knowledge proofs, to maintain transparency
Access Control Mechanisms	Access control is based on roles	Fine-grained access control with multi-signature authentication and time-lock techniques.
Smart Contract Language	Primarily Solidity for Ethereum	Supports multiple languages such as Vyper,Solidity, Michelson,Rust
Gas Efficiency	Inefficient code leading to high transaction costs	Optimized code generating strategies minimize gas use and reduce expenses for consumers.

The MetaMask wallet integration is completed in order to establish a connection with the blockchain and permit transactions.

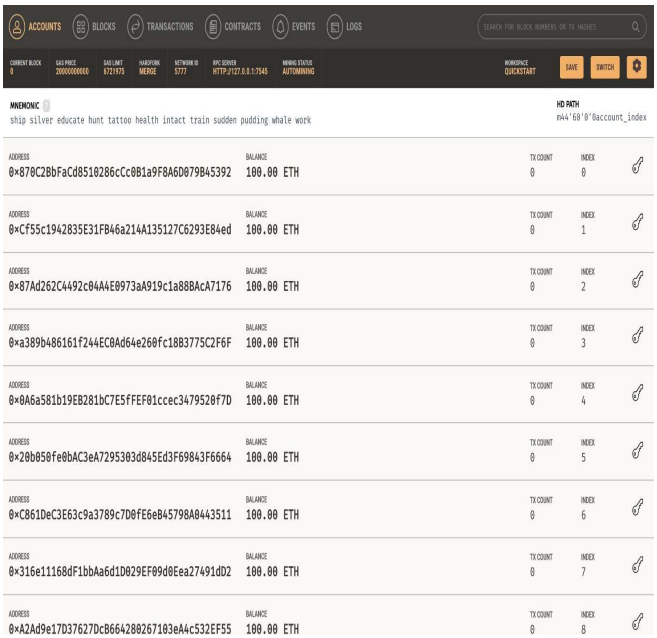

Figure. 4 . Reward collection using Ganache Tool

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The Table 1 shows the comparison between traditional approach and proposed approach. The user may access their account address and use it to communicate with their staking smart contract when they have approved the connection. Figure 4 depicts the reward collection framework of the proposed work. The tool is part of the truffle suit in the planned work Ganache, as shown in Figure above. It is a private Ethereum network that enables the establishment of a regional blockchain ecosystem. Meta Mask wallet is a Chrome browser addon that may be used to make sure transactions can be completed. The following are the functions of the Decentralized Bank created by this contract:

1.Deposit Tokens: Users deposit Tether tokens into the Decentral Bank contract for staking using this function. The function adds the user to the stakes array if they have never staked before, changes the user's staking balance, and modifies the is Staking and has Staked mappings to reflect their current staking status.

2. Unstake Tokens: Users can unstack their previously deposited Tether tokens from the Decentral Bank contract by using this function. The function resets the user's staking balance, sends the staked tokens back to their address, determines if they have any remaining balance, and modifies the is Staking mapping to indicate that they are no longer staking.

3. Issue Tokens: The contract owner uses this function to reward customers who have invested their Tether tokens in the Decentral Bank contract with RWD tokens. The function moves the reward tokens from the contract owner's address to the staker's address after iterating through the array of stakers and calculating the reward amount for each staker.

V. CONCLUSION

The proposed paper presents a unique architecture for decentralized finance that efficiently tackles major problems in the existing financial ecosystem such as transaction process inefficiencies, security flaws, and restricted accessibility. The suggested method not only increases user trust and security but also makes sure that financial services are more accessible to marginalized communities by combining cutting-edge cryptography techniques with creative governance structures. The study emphasizes how emerging technologies like decentralized governance and smart contracts may be used to improve the dependability and transparency of financial operations. These developments increase user involvement and trust while also reducing the hazards that are frequently connected to DeFi. The proposed concept facilitates safe and effective transactions and can be easily adapted and replicated for a variety of industries, which will ultimately lead to a wider use of DeFi solutions. Future research work should investigate the effects of current and proposed rules on DeFi platforms and provide governance structures that can adjust to changes in regulations without compromising the fundamental ideas of user empowerment and decentralization.

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